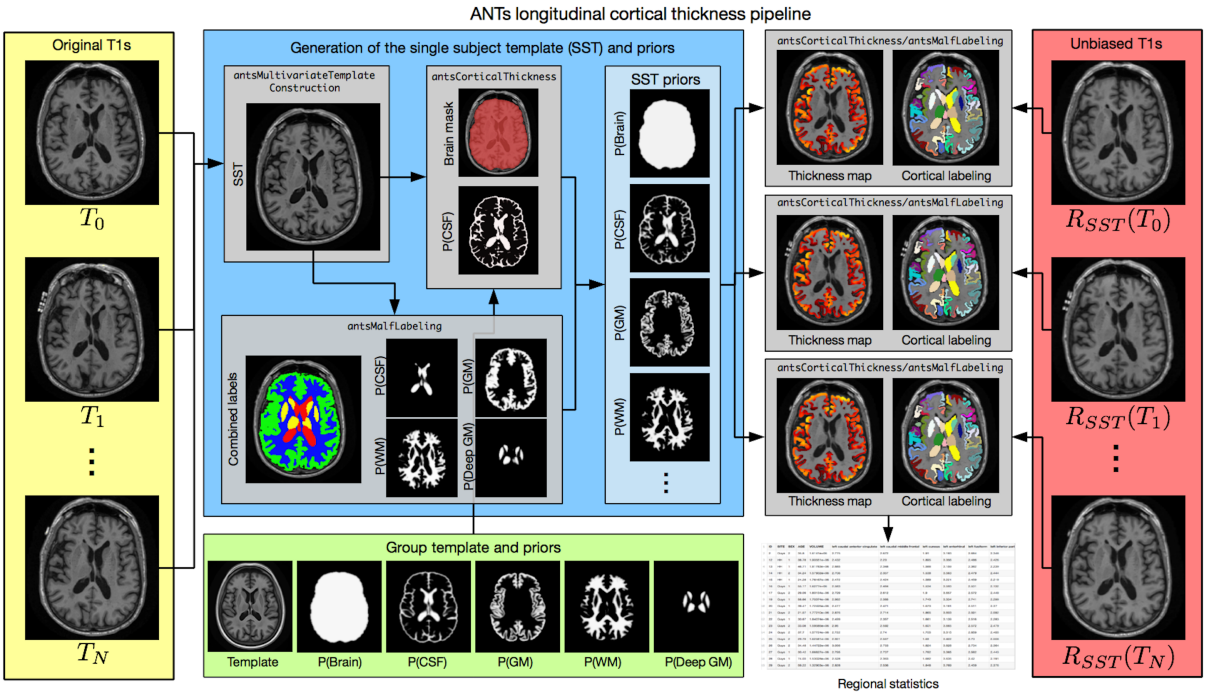


Figure 1: Diagrammatic illustration of the ANTs longitudinal cortical thickness pipeline for a single subject with N time points. From the N original T1-weighted images (left column, yellow panel) and the group template and priors (bottom row, green panel), the single-subject template (SST) and auxiliary prior images are created (center, blue panel). These subject-specific template and other auxiliary images are used to generate the individual time-point cortical thickness maps, in the individual time point's native space (denoted as "ANTs Native" in the text). Optionally, one can rigidly transform the time-point images prior to segmentation and cortical thickness estimation (right column, red panel). This alternative processing scheme is referred to as "ANTs SST". For regional thickness values, regional labels are propagated to each image using a given atlas set (with cortical labels) and joint label fusion.

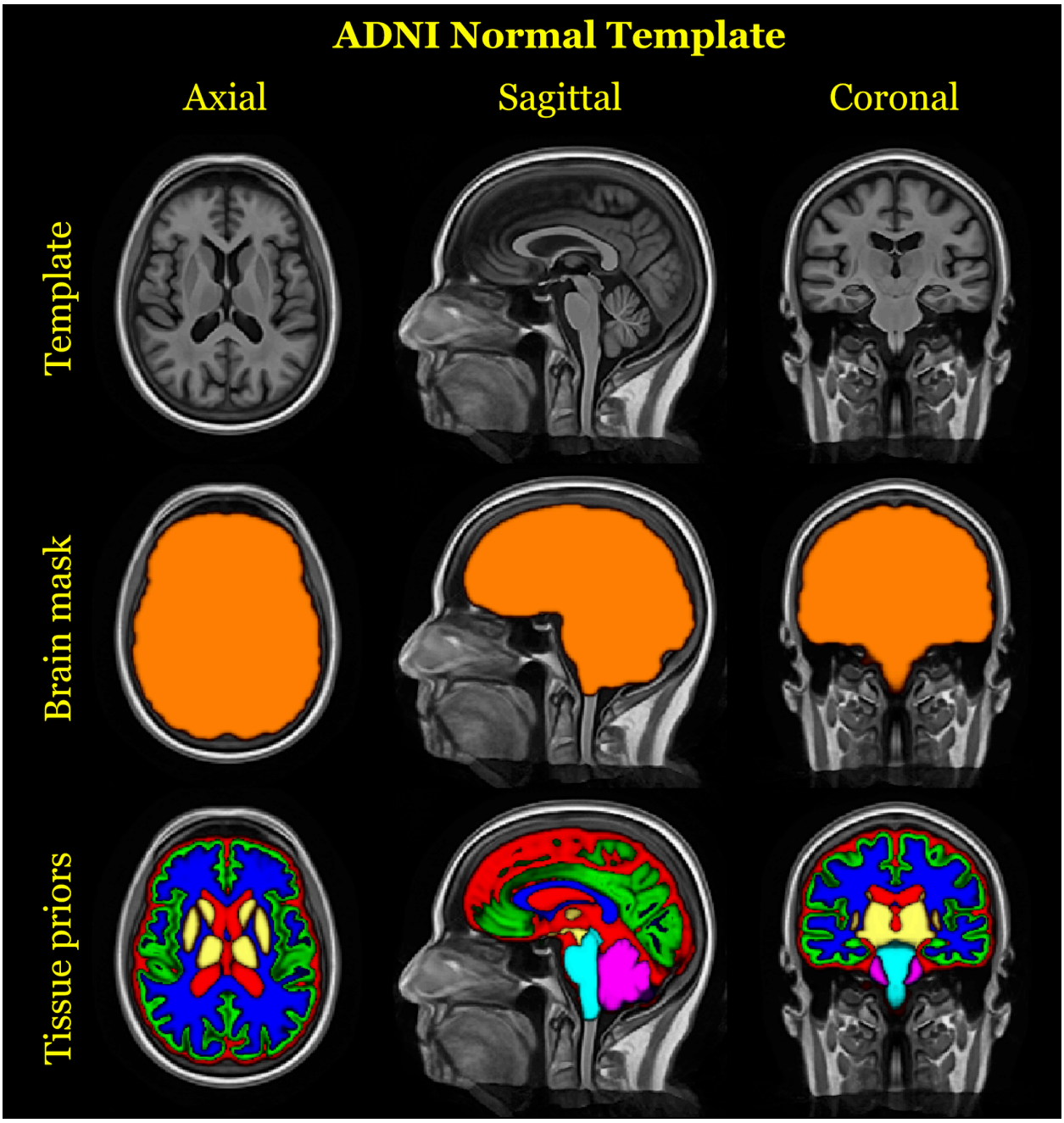


Figure 2: Top row: Canonical views of the template created from 52 randomly selected cognitively normal subjects of the ADNI-1 database. The prior probability mask for the whole brain (middle row) and the six tissue priors (bottom row) are used to "seed" each single-subject template for creation of a probabilistic brain mask and probabilistic tissues priors during longitudinal processing.

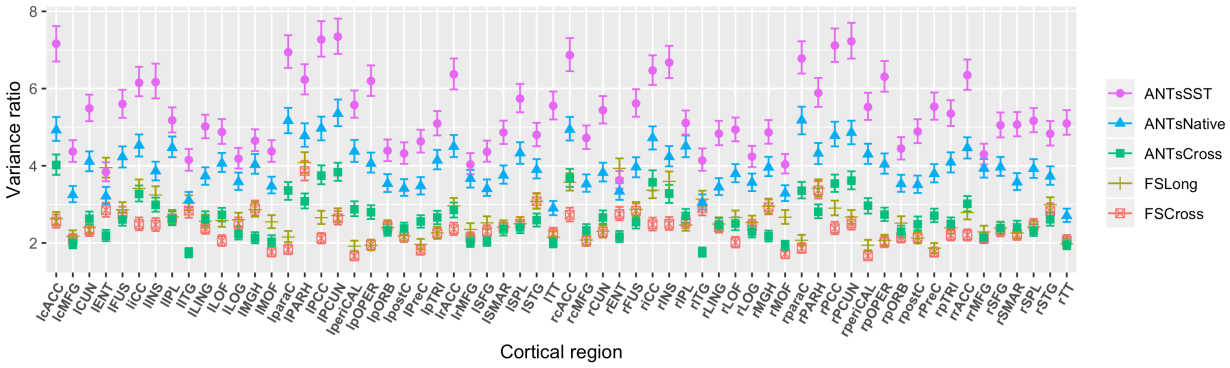


Figure 3: 95% credible intervals of the region-specific variance ratios $r^k = \tau_k/\sigma_k$ are presented for each processing method. The ANTs SST method dominates the others across the majority of regions—its point estimates (posterior medians) are greater than those of the other processing methods except for the left and right EC values in FreeSurfer Long (although there is significant overlap in the credible intervals in those regions). These results also suggest that longitudinal processing is to be preferred for both packages.

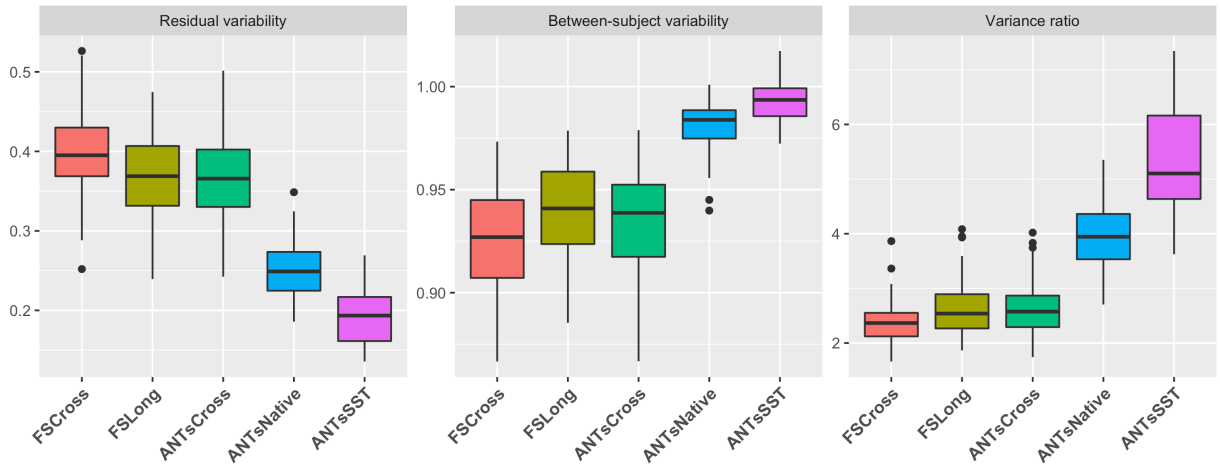


Figure 4: Box plots showing the distribution of the residual variability, between subject variability, and ratio of the between-subject variability and residual variability for each of the 62 DKT regions. Note that the "better" measurement maximizes this latter ratio.

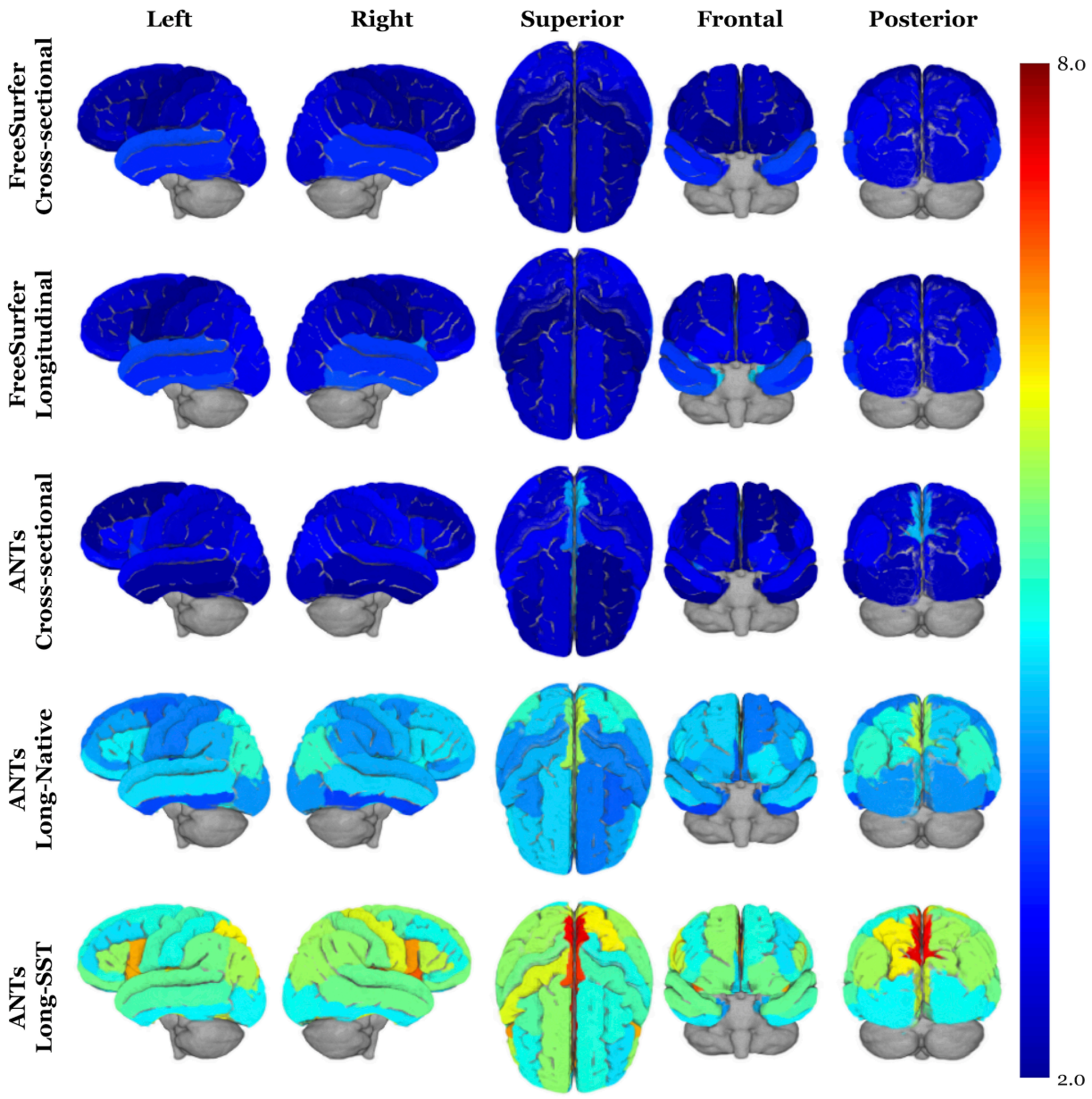


Figure 5: 3-D volumetric rendering of the regional variance ratio values on the generated ADNI template. The higher variance ratios indicate greater between-subject to residual variability.

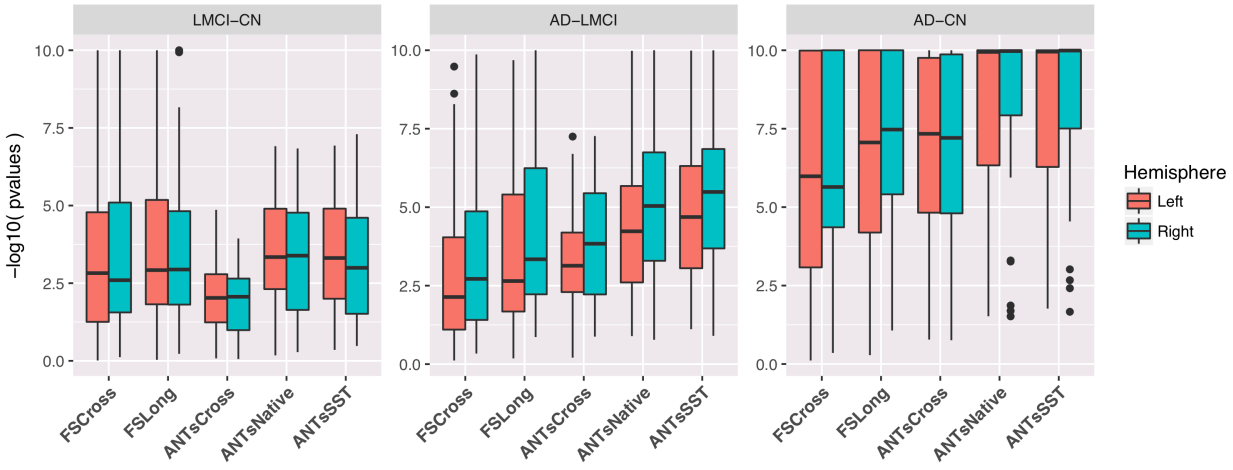


Figure 6: Log-scaled p-values summarizing Tables 2 and 3 demonstrating performance differences across cross-sectional and longitudinal pipelines for the three diagnostic contrasts.

Table 1: The 31 cortical labels (per hemisphere) of the Desikan-Killiany-Tourville atlas. The ROI abbreviations from the R `brainGraph` package are given in parentheses and used in later figures.

1) caudal anterior cingulate (`cACC`)	17) pars orbitalis (`pORB`)
2) caudal middle frontal (`cMFG`)	18) pars triangularis (`pTRI`)
3) cuneus (`CUN`)	19) pericalcarine (`periCAL`)
4) entorhinal (`ENT`)	20) postcentral (`postC`)
5) fusiform (`FUS`)	21) posterior cingulate (`PCC`)
6) inferior parietal (`IPL`)	22) precentral (`preC`)
7) inferior temporal (`ITG`)	23) precuneus (`PCUN`)
8) isthmus cingulate (`iCC`)	24) rosterior anterior cingulate (`rACC`)
9) lateral occipital (`LOG`)	25) rostral middle frontal (`rMFG`)
10) lateral orbitofrontal (`LOF`)	26) superior frontal (`SFG`)
11) lingual (`LING`)	27) superior parietal (`SPL`)
12) medial orbitofrontal (`MOF`)	28) superior temporal (`STG`)
13) middle temporal (`MTG`)	29) supramarginal (`SMAR`)
14) parahippocampal (`PARH`)	30) transverse temporal (`TT`)
15) paracentral (`paraC`)	31) insula (`INS`)
16) pars opercularis (`pOPER`)	

Table 2: 95% confidence intervals for the diagnostic contrasts (LMCI−CN, AD−LMCI, AD−CN) of the ADNI-1 data set for each DKT region of the left hemisphere. Each cell is color-coded based on the adjusted log-scaled p-value significance from dark orange (p < 1e-10) to yellow (p = 0.1). Absence of color denotes nonsignificance.

	LMCI−CN					AD−LMCI					AD−CN				
DKT	FSCross	FSLong	ANTsCross	ANTsNative	ANTsSST	FSCross	FSLong	ANTsCross	ANTsNative	ANTsSST	FSCross	FSLong	ANTsCross	ANTsNative	ANTsSST
lcACC	-0.03,0.068	-0.038,0.058	-0.132,0.024	-0.235,-0.065	-0.238,-0.065	-0.075,0.033	-0.08,0.024	-0.103,0.068	-0.187,-0.001	-0.211,-0.022	-0.063,0.058	-0.077,0.041	-0.168,0.024	-0.349,-0.14	-0.375,-0.162
lcMFG	-0.111,-0.043	-0.11,-0.04	-0.197,-0.067	-0.188,-0.063	-0.188,-0.061	-0.109,-0.034	-0.13,-0.053	-0.201,-0.058	-0.21,-0.073	-0.201,-0.062	-0.19,-0.106	-0.21,-0.124	-0.342,-0.182	-0.345,-0.19	-0.334,-0.178
lCUN	-0.042,0.004	-0.046,0.002	-0.049,0.049	-0.073,0.038	-0.08,0.03	-0.03,0.021	-0.034,0.019	-0.108,0	-0.138,-0.016	-0.145,-0.024	-0.053,0.005	-0.059,0	-0.115,0.006	-0.164,-0.026	-0.177,-0.041
lENT	-0.404,-0.24	-0.405,-0.24	-0.285,-0.054	-0.479,-0.219	-0.486,-0.223	-0.359,-0.179	-0.385,-0.204	-0.354,-0.1	-0.462,-0.177	-0.514,-0.226	-0.692,-0.489	-0.719,-0.515	-0.539,-0.254	-0.83,-0.509	-0.887,-0.562
lFUS	-0.129,-0.059	-0.126,-0.054	-0.196,-0.029	-0.308,-0.116	-0.321,-0.128	-0.149,-0.073	-0.161,-0.082	-0.302,-0.117	-0.398,-0.187	-0.418,-0.207	-0.248,-0.162	-0.255,-0.167	-0.425,-0.218	-0.623,-0.385	-0.656,-0.418
lIPL	-0.099,-0.037	-0.099,-0.035	-0.245,-0.069	-0.245,-0.07	-0.246,-0.067	-0.138,-0.07	-0.144,-0.074	-0.357,-0.163	-0.349,-0.157	-0.344,-0.148	-0.21,-0.134	-0.215,-0.136	-0.526,-0.308	-0.52,-0.302	-0.514,-0.292
lITG	-0.146,-0.07	-0.148,-0.072	-0.225,-0.012	-0.359,-0.132	-0.406,-0.165	-0.165,-0.082	-0.17,-0.086	-0.454,-0.218	-0.535,-0.286	-0.574,-0.31	-0.279,-0.185	-0.285,-0.19	-0.586,-0.322	-0.797,-0.516	-0.876,-0.579
liCC	-0.091,-0.02	-0.096,-0.02	-0.17,-0.025	-0.252,-0.089	-0.255,-0.089	-0.117,-0.039	-0.137,-0.054	-0.231,-0.071	-0.307,-0.129	-0.328,-0.147	-0.177,-0.09	-0.2,-0.107	-0.338,-0.159	-0.489,-0.288	-0.512,-0.307
lLOG	-0.065,-0.013	-0.062,-0.01	-0.172,-0.017	-0.18,-0.027	-0.176,-0.021	-0.07,-0.013	-0.074,-0.017	-0.262,-0.091	-0.285,-0.119	-0.304,-0.134	-0.112,-0.049	-0.113,-0.049	-0.367,-0.175	-0.399,-0.211	-0.413,-0.222
lLOF	-0.062,0	-0.071,-0.011	-0.153,0	-0.23,-0.068	-0.236,-0.069	-0.078,-0.01	-0.086,-0.02	-0.217,-0.048	-0.279,-0.101	-0.31,-0.127	-0.114,-0.037	-0.131,-0.057	-0.303,-0.114	-0.439,-0.239	-0.474,-0.269
lLING	-0.041,0.005	-0.041,0.004	-0.097,0.033	-0.148,-0.003	-0.157,-0.015	-0.065,-0.015	-0.07,-0.02	-0.141,0.003	-0.184,-0.024	-0.19,-0.035	-0.086,-0.03	-0.091,-0.035	-0.181,-0.02	-0.269,-0.09	-0.285,-0.111
lMOF	-0.079,-0.011	-0.095,-0.03	-0.181,-0.004	-0.294,-0.114	-0.303,-0.114	-0.08,-0.005	-0.086,-0.015	-0.255,-0.059	-0.326,-0.128	-0.378,-0.171	-0.13,-0.046	-0.153,-0.073	-0.359,-0.139	-0.542,-0.32	-0.599,-0.367
lMGH	-0.147,-0.076	-0.137,-0.066	-0.239,-0.06	-0.32,-0.142	-0.35,-0.161	-0.166,-0.088	-0.167,-0.089	-0.361,-0.162	-0.442,-0.247	-0.471,-0.265	-0.283,-0.194	-0.274,-0.186	-0.523,-0.3	-0.686,-0.465	-0.74,-0.506
lPARH	-0.151,-0.044	-0.137,-0.036	-0.189,-0.022	-0.26,-0.074	-0.265,-0.077	-0.189,-0.071	-0.182,-0.07	-0.301,-0.116	-0.355,-0.152	-0.37,-0.164	-0.294,-0.161	-0.275,-0.15	-0.417,-0.211	-0.536,-0.306	-0.554,-0.322
lparaC	-0.075,-0.011	-0.088,-0.021	-0.075,0.024	-0.087,0.027	-0.087,0.027	-0.037,0.034	-0.068,0.006	-0.111,-0.002	-0.114,0.011	-0.122,0.003	-0.084,-0.005	-0.127,-0.044	-0.143,-0.021	-0.152,-0.011	-0.16,-0.019
lpOPER	-0.07,-0.007	-0.074,-0.011	-0.188,-0.065	-0.222,-0.099	-0.211,-0.091	-0.072,-0.002	-0.08,-0.01	-0.212,-0.076	-0.218,-0.084	-0.212,-0.08	-0.115,-0.037	-0.127,-0.048	-0.346,-0.194	-0.387,-0.236	-0.371,-0.223
lpORB	-0.105,-0.031	-0.099,-0.026	-0.163,0.007	-0.203,-0.041	-0.205,-0.042	-0.053,0.028	-0.067,0.014	-0.225,-0.037	-0.244,-0.066	-0.267,-0.088	-0.126,-0.034	-0.134,-0.044	-0.314,-0.104	-0.377,-0.176	-0.402,-0.2
lpTRI	-0.094,-0.035	-0.099,-0.038	-0.183,-0.029	-0.209,-0.066	-0.191,-0.046	-0.073,-0.008	-0.083,-0.016	-0.217,-0.047	-0.229,-0.072	-0.24,-0.081	-0.142,-0.068	-0.155,-0.08	-0.333,-0.143	-0.377,-0.2	-0.369,-0.189
lperiCAL	-0.019,0.021	-0.031,0.016	-0.06,0.065	-0.079,0.058	-0.095,0.042	-0.028,0.017	-0.044,0.008	-0.168,-0.03	-0.19,-0.04	-0.207,-0.057	-0.029,0.021	-0.055,0.003	-0.174,-0.019	-0.211,-0.041	-0.243,-0.074
lpostC	-0.05,-0.001	-0.052,-0.001	-0.12,-0.03	-0.109,-0.012	-0.11,-0.009	-0.051,0.004	-0.062,-0.005	-0.13,-0.03	-0.14,-0.033	-0.143,-0.031	-0.08,-0.019	-0.092,-0.028	-0.211,-0.1	-0.207,-0.087	-0.209,-0.084
lPCC	-0.058,0.004	-0.068,-0.003	-0.157,-0.026	-0.258,-0.103	-0.267,-0.104	-0.075,-0.006	-0.084,-0.013	-0.177,-0.033	-0.244,-0.074	-0.272,-0.093	-0.106,-0.029	-0.124,-0.044	-0.278,-0.116	-0.436,-0.244	-0.469,-0.267
lPreC	-0.092,-0.023	-0.101,-0.03	-0.145,-0.055	-0.138,-0.036	-0.135,-0.027	-0.077,-0.001	-0.093,-0.013	-0.141,-0.042	-0.145,-0.032	-0.139,-0.021	-0.139,-0.054	-0.162,-0.074	-0.247,-0.136	-0.239,-0.112	-0.228,-0.095
lPCUN	-0.099,-0.042	-0.105,-0.045	-0.17,-0.041	-0.199,-0.05	-0.197,-0.045	-0.091,-0.028	-0.112,-0.046	-0.208,-0.067	-0.241,-0.078	-0.245,-0.08	-0.165,-0.095	-0.191,-0.117	-0.323,-0.164	-0.376,-0.193	-0.378,-0.19
lrACC	-0.088,0.001	-0.095,-0.011	-0.133,0.05	-0.253,-0.05	-0.249,-0.042	-0.054,0.044	-0.054,0.039	-0.151,0.05	-0.266,-0.044	-0.308,-0.083	-0.103,0.007	-0.113,-0.008	-0.205,0.021	-0.431,-0.182	-0.468,-0.214
lrMFG	-0.087,-0.032	-0.095,-0.039	-0.247,-0.054	-0.293,-0.115	-0.297,-0.108	-0.087,-0.027	-0.089,-0.027	-0.292,-0.078	-0.302,-0.106	-0.332,-0.125	-0.151,-0.083	-0.16,-0.091	-0.455,-0.216	-0.519,-0.298	-0.548,-0.314
lSFG	-0.107,-0.049	-0.112,-0.053	-0.197,-0.076	-0.218,-0.093	-0.215,-0.088	-0.086,-0.023	-0.099,-0.033	-0.202,-0.069	-0.234,-0.098	-0.242,-0.103	-0.168,-0.097	-0.186,-0.112	-0.347,-0.197	-0.398,-0.244	-0.403,-0.246
lSPL	-0.086,-0.026	-0.087,-0.023	-0.143,-0.024	-0.105,0.016	-0.098,0.024	-0.08,-0.014	-0.097,-0.026	-0.164,-0.033	-0.146,-0.014	-0.141,-0.008	-0.14,-0.066	-0.156,-0.077	-0.256,-0.109	-0.199,-0.05	-0.187,-0.037
lSTG	-0.137,-0.069	-0.132,-0.066	-0.201,-0.074	-0.228,-0.105	-0.23,-0.105	-0.132,-0.057	-0.133,-0.06	-0.244,-0.104	-0.289,-0.153	-0.297,-0.161	-0.239,-0.155	-0.237,-0.155	-0.39,-0.233	-0.464,-0.312	-0.474,-0.319
lSMAR	-0.09,-0.026	-0.091,-0.027	-0.209,-0.074	-0.194,-0.064	-0.195,-0.058	-0.109,-0.038	-0.122,-0.051	-0.234,-0.086	-0.236,-0.094	-0.24,-0.09	-0.171,-0.092	-0.185,-0.106	-0.385,-0.219	-0.374,-0.214	-0.376,-0.208
lTT	-0.09,0.003	-0.088,0.001	-0.112,-0.002	-0.118,-0.023	-0.103,-0.014	-0.082,0.02	-0.091,0.007	-0.16,-0.038	-0.142,-0.037	-0.137,-0.04	-0.132,-0.016	-0.141,-0.031	-0.224,-0.088	-0.218,-0.101	-0.201,-0.092
lINS	-0.097,-0.023	-0.098,-0.023	-0.208,-0.045	-0.275,-0.108	-0.274,-0.109	-0.089,-0.007	-0.102,-0.02	-0.248,-0.069	-0.349,-0.165	-0.351,-0.171	-0.153,-0.062	-0.167,-0.075	-0.386,-0.185	-0.552,-0.345	-0.554,-0.351

Table 3: 95% confidence intervals for the diagnostic contrasts (LMCI−CN, AD−LMCI, AD−CN) of the ADNI-1 data set for each DKT region of the right hemisphere. Each cell is color-coded based on the adjusted log-scaled p-value significance from dark orange (p < 1e-10) to yellow (p = 0.1). Absence of color denotes nonsignificance.

	LMCI−CN					AD−LMCI					AD−CN				
DKT	FSCross	FSLong	ANTsCross	ANTsNative	ANTsSST	FSCross	FSLong	ANTsCross	ANTsNative	ANTsSST	FSCross	FSLong	ANTsCross	ANTsNative	ANTsSST
rcACC	-0.058,0.024	-0.048,0.028	-0.138,0.004	-0.214,-0.066	-0.222,-0.072	-0.088,0.002	-0.076,0.008	-0.141,0.015	-0.203,-0.041	-0.214,-0.05	-0.11,-0.009	-0.091,0.003	-0.217,-0.043	-0.353,-0.171	-0.371,-0.187
rcMFG	-0.117,-0.048	-0.114,-0.042	-0.184,-0.059	-0.178,-0.053	-0.178,-0.05	-0.118,-0.041	-0.14,-0.061	-0.248,-0.11	-0.251,-0.114	-0.241,-0.101	-0.205,-0.119	-0.223,-0.134	-0.377,-0.223	-0.376,-0.221	-0.364,-0.206
rCUN	-0.025,0.022	-0.03,0.019	-0.059,0.042	-0.085,0.032	-0.094,0.021	-0.057,-0.006	-0.062,-0.008	-0.12,-0.008	-0.133,-0.004	-0.14,-0.015	-0.062,-0.004	-0.071,-0.01	-0.135,-0.01	-0.167,-0.023	-0.184,-0.044
rENT	-0.408,-0.254	-0.403,-0.247	-0.271,-0.026	-0.446,-0.201	-0.438,-0.184	-0.331,-0.162	-0.356,-0.187	-0.407,-0.136	-0.511,-0.242	-0.561,-0.282	-0.672,-0.482	-0.692,-0.5	-0.571,-0.269	-0.852,-0.548	-0.89,-0.576
rFUS	-0.114,-0.045	-0.114,-0.043	-0.221,-0.046	-0.337,-0.144	-0.343,-0.15	-0.149,-0.073	-0.165,-0.086	-0.347,-0.154	-0.44,-0.228	-0.462,-0.252	-0.233,-0.148	-0.249,-0.16	-0.492,-0.276	-0.694,-0.455	-0.722,-0.485
rIPL	-0.107,-0.042	-0.113,-0.045	-0.201,-0.025	-0.228,-0.057	-0.229,-0.051	-0.15,-0.079	-0.16,-0.086	-0.366,-0.172	-0.356,-0.169	-0.358,-0.164	-0.229,-0.149	-0.243,-0.16	-0.491,-0.274	-0.51,-0.299	-0.512,-0.291
rITG	-0.162,-0.086	-0.14,-0.065	-0.281,-0.064	-0.408,-0.19	-0.44,-0.208	-0.179,-0.096	-0.193,-0.111	-0.432,-0.191	-0.539,-0.299	-0.591,-0.337	-0.308,-0.215	-0.301,-0.208	-0.618,-0.35	-0.853,-0.583	-0.931,-0.645
riCC	-0.08,-0.007	-0.084,-0.005	-0.182,-0.044	-0.256,-0.097	-0.26,-0.099	-0.116,-0.035	-0.139,-0.053	-0.239,-0.087	-0.319,-0.145	-0.335,-0.16	-0.164,-0.073	-0.189,-0.092	-0.361,-0.19	-0.507,-0.31	-0.526,-0.328
rLOG	-0.055,-0.004	-0.055,-0.004	-0.152,0.016	-0.179,-0.018	-0.184,-0.017	-0.073,-0.018	-0.08,-0.024	-0.31,-0.124	-0.325,-0.148	-0.337,-0.155	-0.106,-0.044	-0.113,-0.05	-0.389,-0.181	-0.434,-0.235	-0.449,-0.244
rLOF	-0.081,-0.018	-0.08,-0.022	-0.11,0.036	-0.192,-0.031	-0.193,-0.026	-0.077,-0.008	-0.09,-0.025	-0.195,-0.035	-0.255,-0.078	-0.286,-0.102	-0.13,-0.053	-0.145,-0.072	-0.241,-0.063	-0.377,-0.179	-0.406,-0.201
rLING	-0.05,-0.006	-0.055,-0.01	-0.087,0.05	-0.153,0.001	-0.16,-0.011	-0.063,-0.014	-0.067,-0.017	-0.14,0.012	-0.18,-0.011	-0.196,-0.033	-0.094,-0.039	-0.103,-0.046	-0.168,0.002	-0.266,-0.077	-0.292,-0.109
rMOF	-0.077,-0.012	-0.08,-0.017	-0.158,0.019	-0.258,-0.077	-0.268,-0.076	-0.088,-0.016	-0.096,-0.027	-0.25,-0.054	-0.318,-0.118	-0.37,-0.16	-0.137,-0.057	-0.149,-0.071	-0.331,-0.112	-0.498,-0.273	-0.555,-0.318
rMGH	-0.162,-0.093	-0.153,-0.084	-0.242,-0.044	-0.331,-0.139	-0.36,-0.149	-0.167,-0.09	-0.17,-0.094	-0.399,-0.18	-0.496,-0.286	-0.526,-0.296	-0.299,-0.213	-0.293,-0.208	-0.555,-0.31	-0.745,-0.507	-0.796,-0.535
rPARH	-0.193,-0.068	-0.184,-0.066	-0.178,-0.005	-0.277,-0.09	-0.267,-0.082	-0.242,-0.106	-0.23,-0.102	-0.324,-0.134	-0.386,-0.182	-0.401,-0.199	-0.382,-0.228	-0.364,-0.219	-0.427,-0.214	-0.583,-0.352	-0.589,-0.36
rparaC	-0.073,-0.006	-0.091,-0.02	-0.091,0.01	-0.093,0.022	-0.094,0.023	-0.051,0.023	-0.075,0.005	-0.101,0.009	-0.111,0.016	-0.117,0.011	-0.095,-0.012	-0.135,-0.046	-0.149,-0.024	-0.154,-0.011	-0.16,-0.016
rpOPER	-0.078,-0.015	-0.078,-0.016	-0.175,-0.05	-0.195,-0.071	-0.192,-0.07	-0.07,0	-0.084,-0.015	-0.188,-0.05	-0.219,-0.083	-0.217,-0.084	-0.12,-0.042	-0.135,-0.058	-0.309,-0.155	-0.36,-0.208	-0.356,-0.207
rpORB	-0.068,0.008	-0.071,0.001	-0.159,-0.006	-0.174,-0.028	-0.149,-0.002	-0.097,-0.013	-0.105,-0.026	-0.237,-0.067	-0.263,-0.103	-0.274,-0.113	-0.132,-0.038	-0.145,-0.056	-0.33,-0.14	-0.373,-0.193	-0.359,-0.179
rpTRI	-0.085,-0.025	-0.083,-0.022	-0.19,-0.044	-0.191,-0.057	-0.184,-0.049	-0.069,-0.002	-0.083,-0.016	-0.232,-0.072	-0.238,-0.092	-0.25,-0.103	-0.128,-0.053	-0.14,-0.064	-0.359,-0.179	-0.372,-0.207	-0.376,-0.21
rperiCAL	-0.022,0.02	-0.035,0.011	-0.052,0.065	-0.09,0.048	-0.11,0.027	-0.032,0.014	-0.046,0.005	-0.124,0.005	-0.16,-0.009	-0.18,-0.03	-0.036,0.016	-0.061,-0.004	-0.125,0.019	-0.19,-0.02	-0.231,-0.062
rpostC	-0.058,-0.009	-0.06,-0.01	-0.118,-0.028	-0.106,-0.007	-0.106,0.002	-0.072,-0.018	-0.079,-0.024	-0.131,-0.032	-0.15,-0.042	-0.165,-0.047	-0.109,-0.048	-0.117,-0.055	-0.21,-0.099	-0.214,-0.092	-0.224,-0.092
rPCC	-0.055,0.005	-0.068,-0.007	-0.153,-0.03	-0.233,-0.085	-0.243,-0.09	-0.068,-0.002	-0.081,-0.013	-0.135,0	-0.202,-0.04	-0.225,-0.058	-0.097,-0.022	-0.123,-0.047	-0.235,-0.083	-0.372,-0.189	-0.402,-0.213
rPreC	-0.094,-0.025	-0.1,-0.029	-0.12,-0.032	-0.113,-0.011	-0.11,0	-0.08,-0.003	-0.1,-0.021	-0.165,-0.068	-0.186,-0.073	-0.196,-0.075	-0.144,-0.057	-0.169,-0.081	-0.247,-0.138	-0.254,-0.128	-0.258,-0.122
rPCUN	-0.102,-0.038	-0.111,-0.044	-0.184,-0.042	-0.21,-0.048	-0.206,-0.042	-0.096,-0.026	-0.119,-0.045	-0.239,-0.084	-0.273,-0.097	-0.285,-0.107	-0.17,-0.092	-0.201,-0.118	-0.362,-0.187	-0.414,-0.214	-0.421,-0.219
rrACC	-0.062,0.024	-0.069,0.012	-0.21,-0.035	-0.336,-0.137	-0.352,-0.149	-0.096,-0.001	-0.096,-0.006	-0.231,-0.039	-0.324,-0.106	-0.362,-0.14	-0.121,-0.014	-0.13,-0.029	-0.365,-0.149	-0.574,-0.329	-0.627,-0.376
rrMFG	-0.096,-0.038	-0.097,-0.037	-0.203,-0.005	-0.238,-0.059	-0.237,-0.049	-0.09,-0.025	-0.1,-0.033	-0.373,-0.154	-0.359,-0.163	-0.381,-0.176	-0.161,-0.089	-0.171,-0.096	-0.49,-0.245	-0.52,-0.299	-0.538,-0.306
rSFG	-0.102,-0.041	-0.107,-0.043	-0.185,-0.062	-0.203,-0.079	-0.198,-0.073	-0.101,-0.033	-0.115,-0.044	-0.217,-0.081	-0.231,-0.095	-0.242,-0.104	-0.177,-0.101	-0.195,-0.115	-0.348,-0.196	-0.38,-0.227	-0.386,-0.231
rSPL	-0.093,-0.031	-0.098,-0.032	-0.141,-0.011	-0.116,0.02	-0.105,0.034	-0.093,-0.024	-0.106,-0.034	-0.207,-0.064	-0.18,-0.03	-0.171,-0.019	-0.158,-0.082	-0.175,-0.094	-0.291,-0.131	-0.237,-0.069	-0.216,-0.045
rSTG	-0.139,-0.071	-0.131,-0.065	-0.184,-0.04	-0.22,-0.087	-0.232,-0.091	-0.132,-0.056	-0.137,-0.064	-0.269,-0.111	-0.295,-0.15	-0.31,-0.155	-0.241,-0.157	-0.24,-0.158	-0.391,-0.213	-0.458,-0.294	-0.482,-0.307
rSMAR	-0.111,-0.049	-0.11,-0.048	-0.214,-0.069	-0.195,-0.061	-0.186,-0.041	-0.117,-0.049	-0.136,-0.067	-0.248,-0.089	-0.27,-0.123	-0.289,-0.131	-0.201,-0.124	-0.22,-0.142	-0.399,-0.221	-0.407,-0.241	-0.412,-0.234
rTT	-0.105,-0.017	-0.1,-0.015	-0.107,0.013	-0.11,-0.006	-0.108,-0.008	-0.105,-0.007	-0.104,-0.01	-0.225,-0.09	-0.192,-0.077	-0.169,-0.06	-0.171,-0.062	-0.167,-0.062	-0.28,-0.129	-0.258,-0.128	-0.235,-0.111
rINS	-0.079,-0.006	-0.083,-0.014	-0.212,-0.051	-0.271,-0.106	-0.272,-0.109	-0.094,-0.014	-0.098,-0.022	-0.22,-0.043	-0.32,-0.139	-0.323,-0.145	-0.141,-0.052	-0.152,-0.066	-0.362,-0.164	-0.52,-0.316	-0.525,-0.324